Date: _____

# Ghazi University

## CS and IT department

**Subject:** programming fundamentals

**Topic:** PFLAB-2

**Section:** BSCS-C

**Submitted by:** Misbah Rehman

**Roll No:** 27

**Submitted To:** Sir Asad Majeed

**Date:** 09-10-2026

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Date:

# TASK-1:

Type and solve the following program in visual studio Run the program and observe their output.

⇒ Input

```
# include <iostream>
using namespace std;
 int main( ) {
  char letter;
  letter = 'A';
cout << letter<< '\n';
  letter = 'B';
 cout << letter<< '\n';
  cin >> letter;
    return 0;
       }
```

output

```
A
B
```

⇒ Input

```
# include <iostream>
using namespace std;
```

output

```
A$
B$
```

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```
int main( ) {
 char letter;
  letter= 65;
cout << letter<< '\n';
  letter= 66;
cout << letter<< '\n';
   cin >> letter;
   return 0;
     }
```

⇒ Input

output

```
#include <iostream>
using namespace std;
int main ( ) {
 int  letter;
  letter = A;
cout << letter<< '\n';
 letter= B;
 cout<< letter<< '\n';
 cin >> letter;
  return 0;
     }
```

65
66

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## TASK-2:

Take two integers as input from user apply arithmetic operator. As print on screen.

Inputs

```
# include <iostream>
using namespace std;
int main ( ) {
   int a,b;
cout << "enter two integers: ";
   cin >> a >> b;
cout<< "Addition: "<<a+b<<endl;
cout<< "subtract: "<< a-b<<endl;
cout<< "multiply:"<< a*b<<endl;
cout << "divide: "<< a/b<<endl;
cout << "Modulus: "<< a% b<< endl;
      return 0;
      }
```

outputs

```
Enter first nmbr:-
   10
Enter second nmbr
   5
Sum = 15
subtract = 5
multiply = 50
Divide = 2
modulus = 0.8
```

## TASK-3:

write output of following code:

```
# include <iostream>
using namespace std;
int main ( ) {
```

output

10

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```
int val1 ;
int val2;
int val3;
cout<<"please enter 3 nmber: ";
cin>> val1 >> val2 >> val3;
cout << val1<< "\n"<< val2 << "\n"<<val3<< "\n";
cout<<val3<< "\n"<<val2<< "\n"<<val1<< "\n";
   system("pause" );
     return 0;
}
```

20
30
30
20
10

TASK-4: write a program calculate the distance between two points. $x_1, y_1, x_2, y_2$ are dable

Formula:

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Inputs

outputs

```
#include <iostream>
#include <Cmath>
using namespace std;
int main() {
double x1, y1, x2, y2, d;
cout<<"enter x1, y1, x2, y2: ";
cin >> x1 >> y1 >> x2 >> y2;
d = sqrt(pow(x2-x1,2)+pow(y2-y1,2));
cout<<" distance = "<< d<< endl;
```

Enter $x_2, y_3, x_5$
$y_7$
Distance = 0S

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```
return 0;
}
```

## TASK-5: Calculate Area of circle

Input

```
#include<iostream>
using namespace std;
int main ( ){
float r, area circle;
const float pI = 3.14159;
cout << "Enter radius:";
cin >> r;
area circle = pI * r * r;
cout<<"Area of circle= "<< areacircle<<endl;
return 0;
}
```

output

```
Enter radiu: 5
Area of
circle : 78.54
```

## ⇒calculate area of Rectangle

Input

```
#include<iostream>
using namespace std;
int main ( ){
float length, width;
cout<<"Enter length & width!";
cin >>length >> width;
cout<<" Area of Rectangle = "<<l*w<<endl;
return 0;
```

output

```
enter length and w
4, 6;
Area of Rectan
24
```

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=>Calculate Area of Triangle

Input

```
#include<iostream>
using namespace std;
int main( ){
float base, height;
cout<<"enter base and height: ";
cin>>base >> height;
cout<<"Area of triangle = "<<base*height*0.5<<endl;
    return 0;
}
```

output

```
enter base and height:
4 6
Area of triangle
12
```

TASK-6: Total purchase a customer in a store is purchasing five items. The price of five items are;

Input

```
#include<iostream>
using namespace std;
int main( ){
double P1=12.95$, P2=24.95, P3=6.95, P4=14.95, P5=3.95;
double subtotal, salesTax, total;
subtotal = P1+P2+P3+P4+P5;
Sales TAX = subtotal *0.06;
total = Subtotal + Sales Tax;
cout<<"price1: $"<<price1<<endl;
cout<<price2: $"<<price2<<endl;
cout<<price3: $"<<price3<<endl;
```

output

```
P1 = $12.95
P2 = $24.95
Price 3 = $6.95
price 4 = $14.95
Price 5 = $3.95
subtotal = $63.7
```

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```
cout<<"price 4: "$"<<price4<<endl;
cout<<"price 5: $"<<price5<<endl;
cout<<"Subtotal:$"<<subtotal<<endl;
cout<<"Sales TAX(6%):$"<<salestax<<endl;
cout<<"Total: $"<<total<<endl;
    return 0;
    }
```

Sales tax (6%): $3.82
Total = $67.575

## BONUS TASK: Break number

```
#include<iostream>
using namespace std;
int main( ) {
int num;
cout<<"enter five-digit integer: ";
cin >> num;
int d1 = num /10000;
int d2 = (num/1000)%10;
int d3= (num/100)% 10;
int d4= (num/10)% 10;
int d5= num % 10;
cout<<d1<<" "<<d2<<" "<<d3<<" "<<d4<<" "<<d5<<" "<<endl;
    return 0;
    }
```

output

enter five-digit integer = 6000

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# HOME TASK - 1: Swap Variable

```
#include <iostream>
using namespace std;
int main( ) {
int var1, var2, temp;
cout <<"enter var1 and var2: ";
cin >> var1 >> var2;
cout <<" before swap: var1 = 10 \n";
cout << before swap: var2 = 20 \n";
 temp = var1;
 var1 = var2;
 var2 = temp;
cout <<"after swap: var1 = 20 \n";
cout << "after swap: var2 = 10";
    return 0;
    }
```

## output:

enter var1 and var2: 10 20
before swap: var1 = 10
before swap: var2 = 20
after swap: var1 = 20
after swap: var2 = 10

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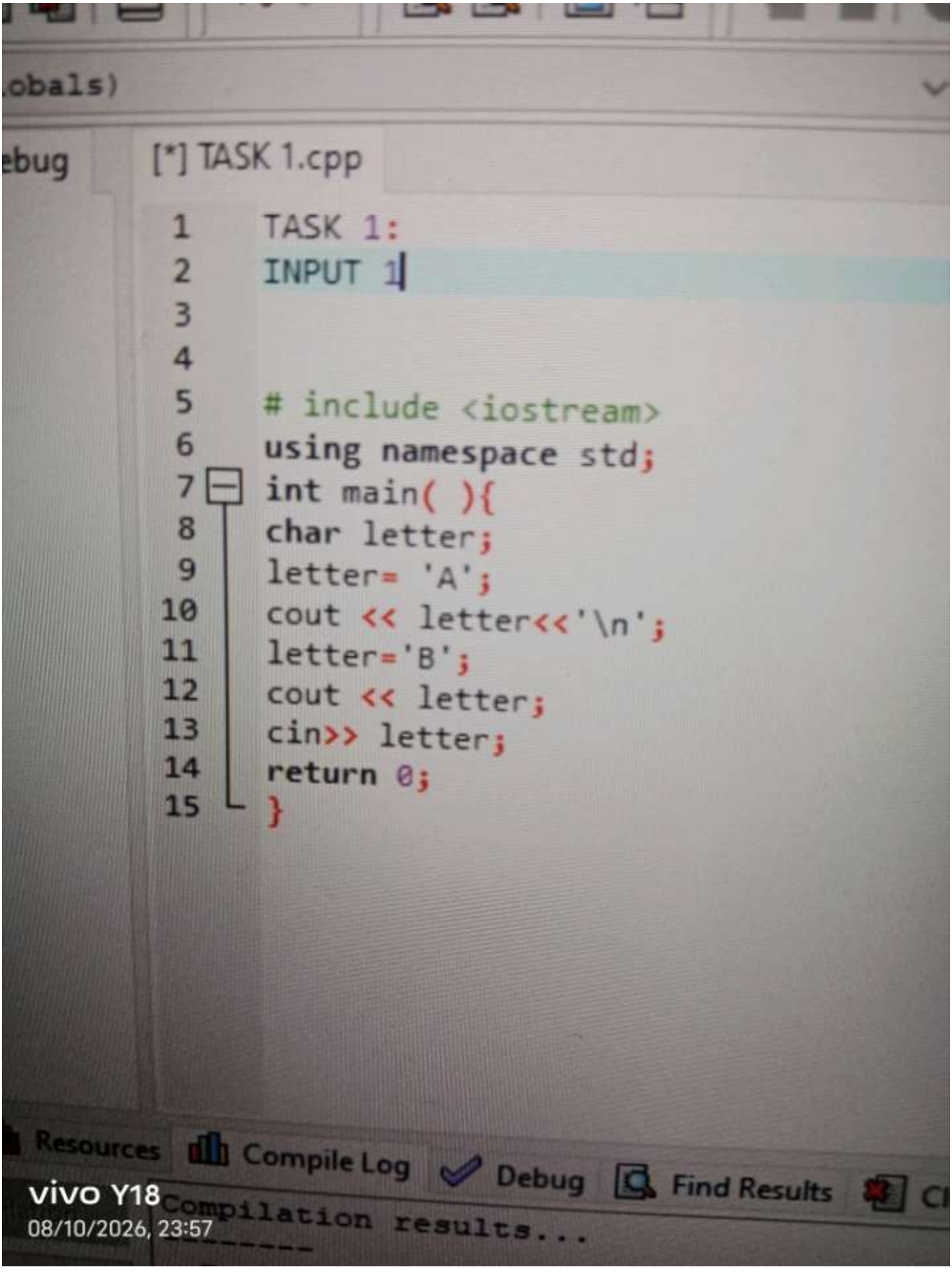
# HOME-TASK: 2: Small To capital

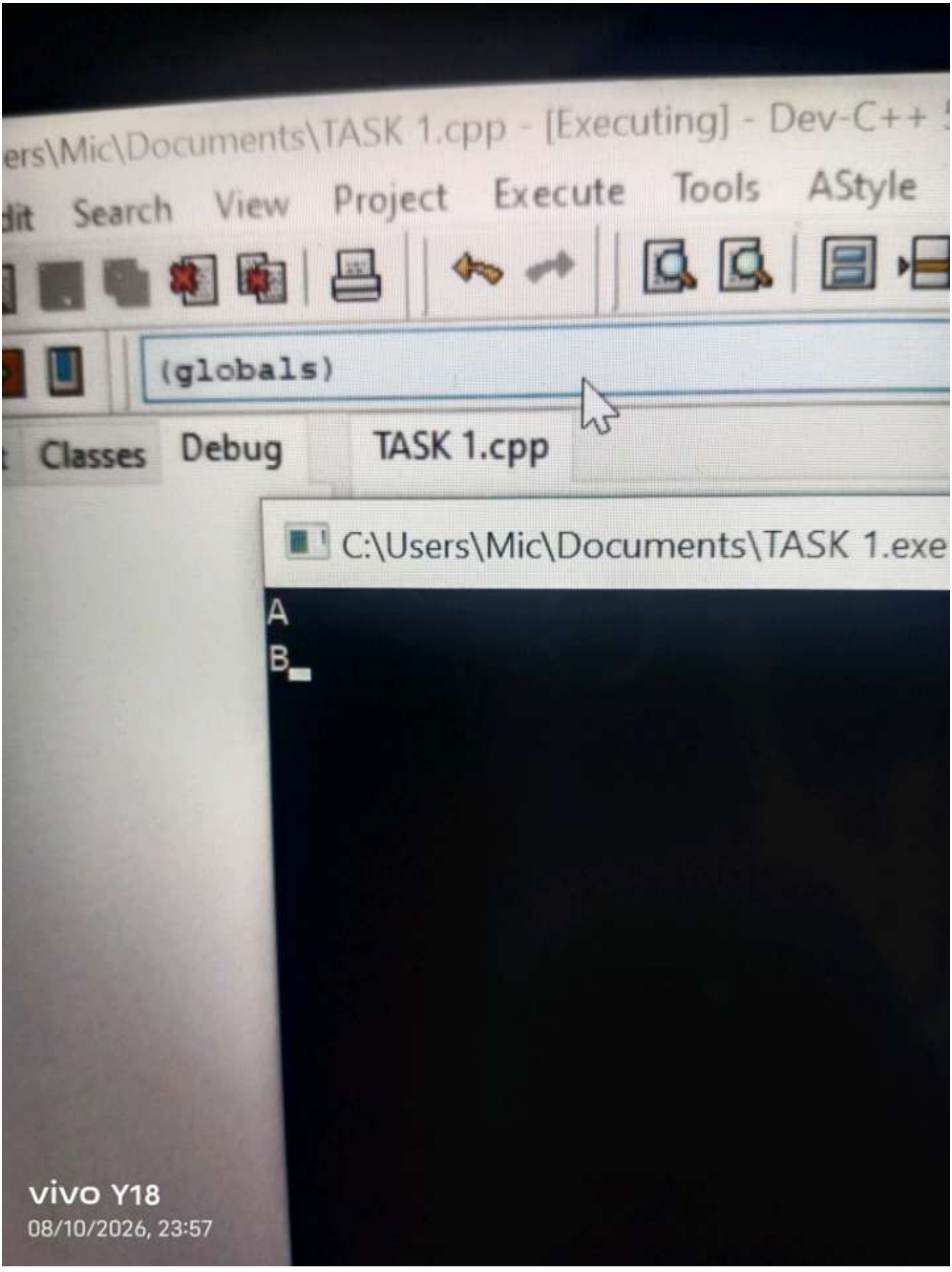
```
# include <iostream>
using namespace std;
int main ( ) {
Char Small;
Cout << "enter Small alphabet (a-z): ";
 Cin >> Small;
char capital = Small - 32;
 Cout << " upper case: \n";
   return 0;
     }
```

## output:

enter small alphabet (a-z): m
upper case:

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obals)
ebug
[*] TASK 1.cpp
1 TASK 1:
2 INPUT 1
3
4
5 # include <iostream>
6 using namespace std;
7 int main( ){
8 char letter;
9 letter= 'A';
10 cout << letter<<'\n';
11 letter='B';
12 cout << letter;
13 cin>> letter;
14 return 0;
15 }
Resources
Compile Log
Debug
Find Results
Compilation results...
vivo Y18
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ers\Mic\Documents\TASK 1.cpp - [Executing] - Dev-C++
dit Search View Project Execute Tools AStyle
(globals)
Classes Debug
TASK 1.cpp
C:\Users\Mic\Documents\TASK 1.exe
A
B
vivo Y18
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obals)

bug

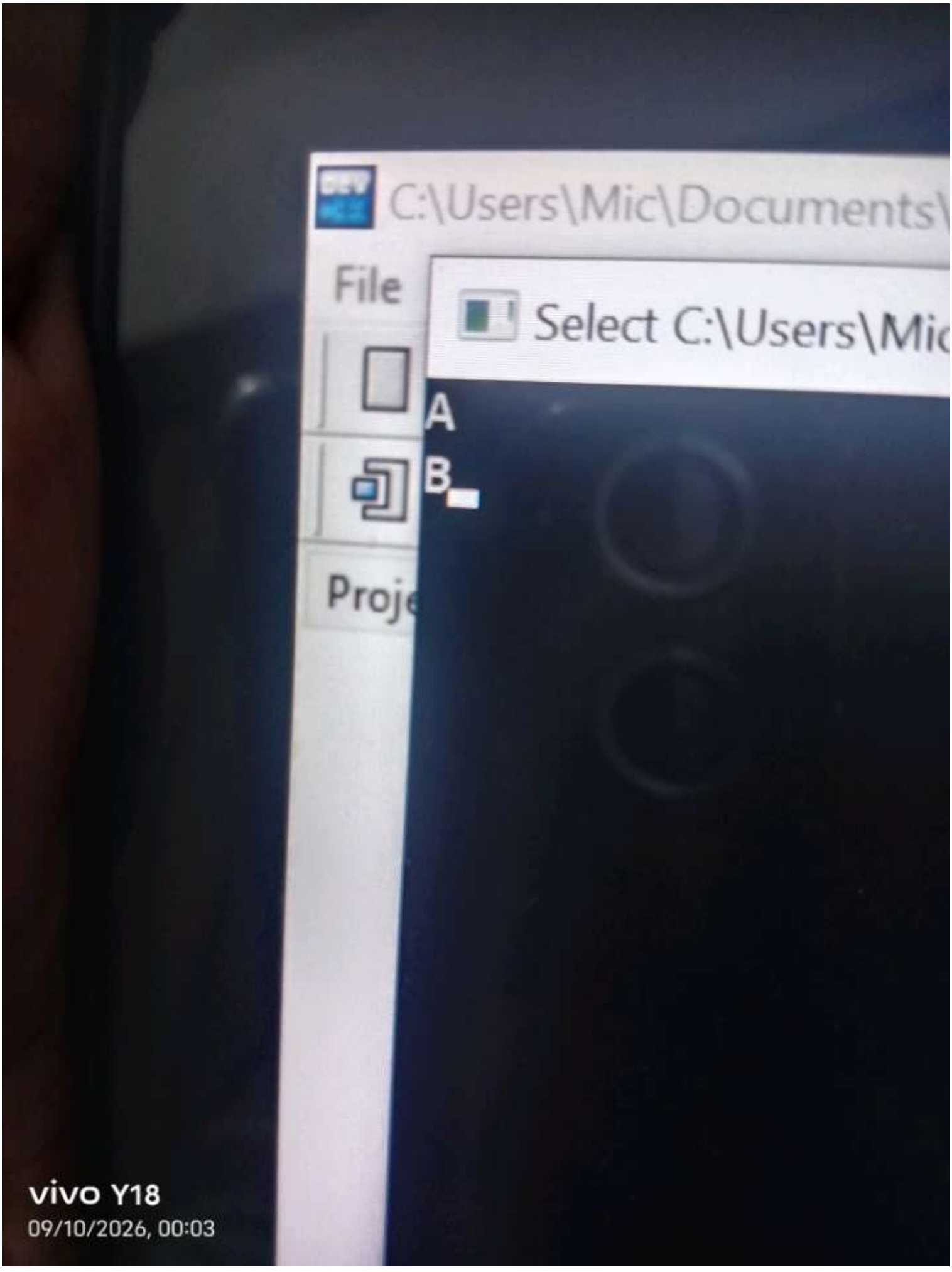
[*] TASK 1.cpp

```
1    TASK 1:
2    INPUT 2
3
4
5    # include <iostream>
6    using namespace std;
7    int main( ){
8    char letter;
9    letter= 65;
10   cout << letter<<'\n';
11   letter= 66;
12   cout << letter;
13   cin>> letter;
14   return 0;
15   }
```

Resources | Compile Log | Debug | Find Results | Cl

Compilation results...

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[*] TASK 1.cpp

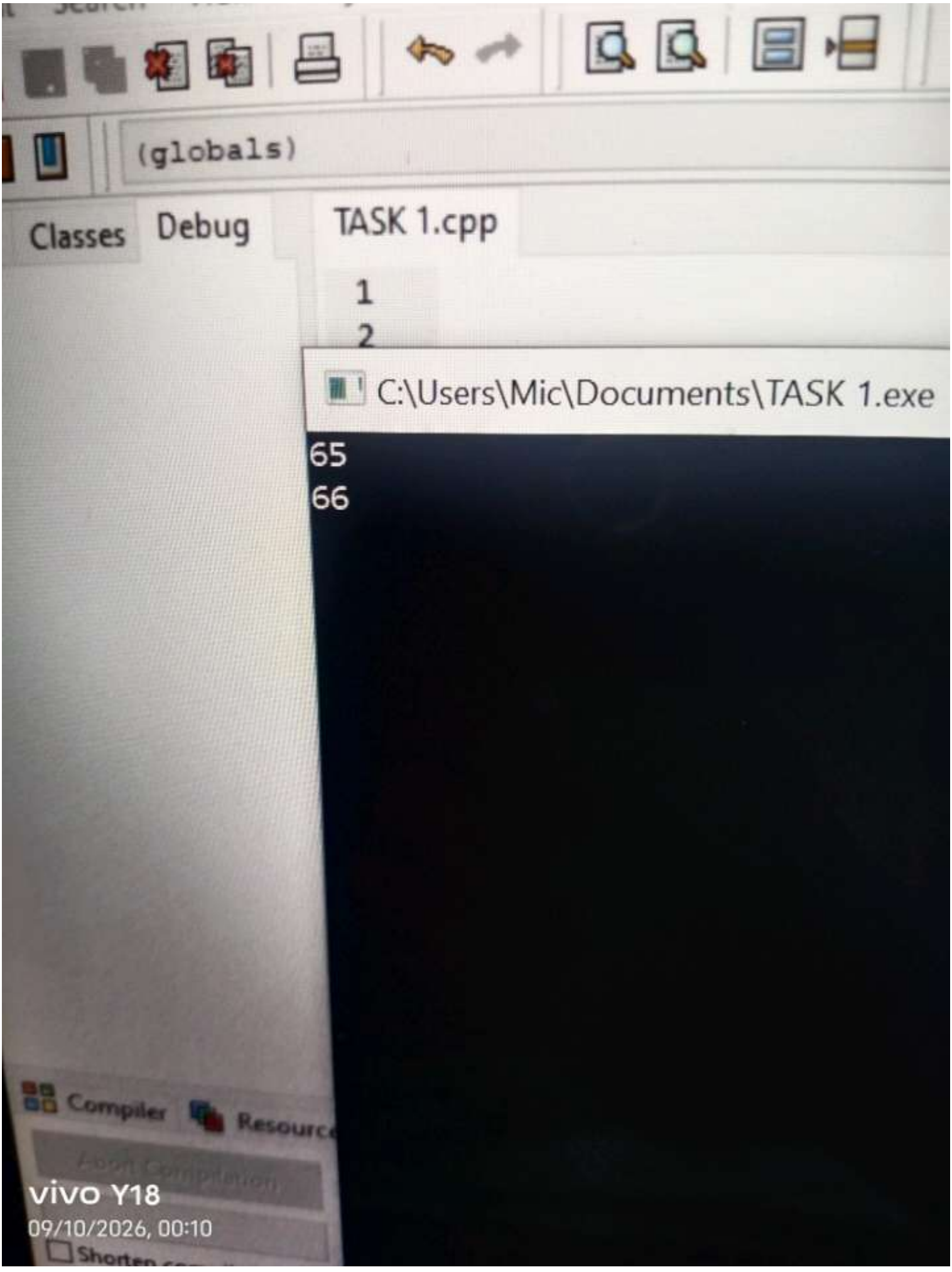
```
1   TASK 1:
2   INPUT 3
3
4
5   # include <iostream>
6   using namespace std;
7   int main( ){
8   int letter;
9   letter= 65;
10  cout << letter<<'\n';
11  letter= 66;
12  cout << letter;
13  cin>> letter;
14  return 0;
15  }
```

Resources | Compile Log | Debug | Find Results

Compilation results...

Errors: 0

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(globals)
Classes
Debug
TASK 1.cpp
1
2
C:\Users\Mic\Documents\TASK 1.exe
65
66
Compiler
vivo Y18
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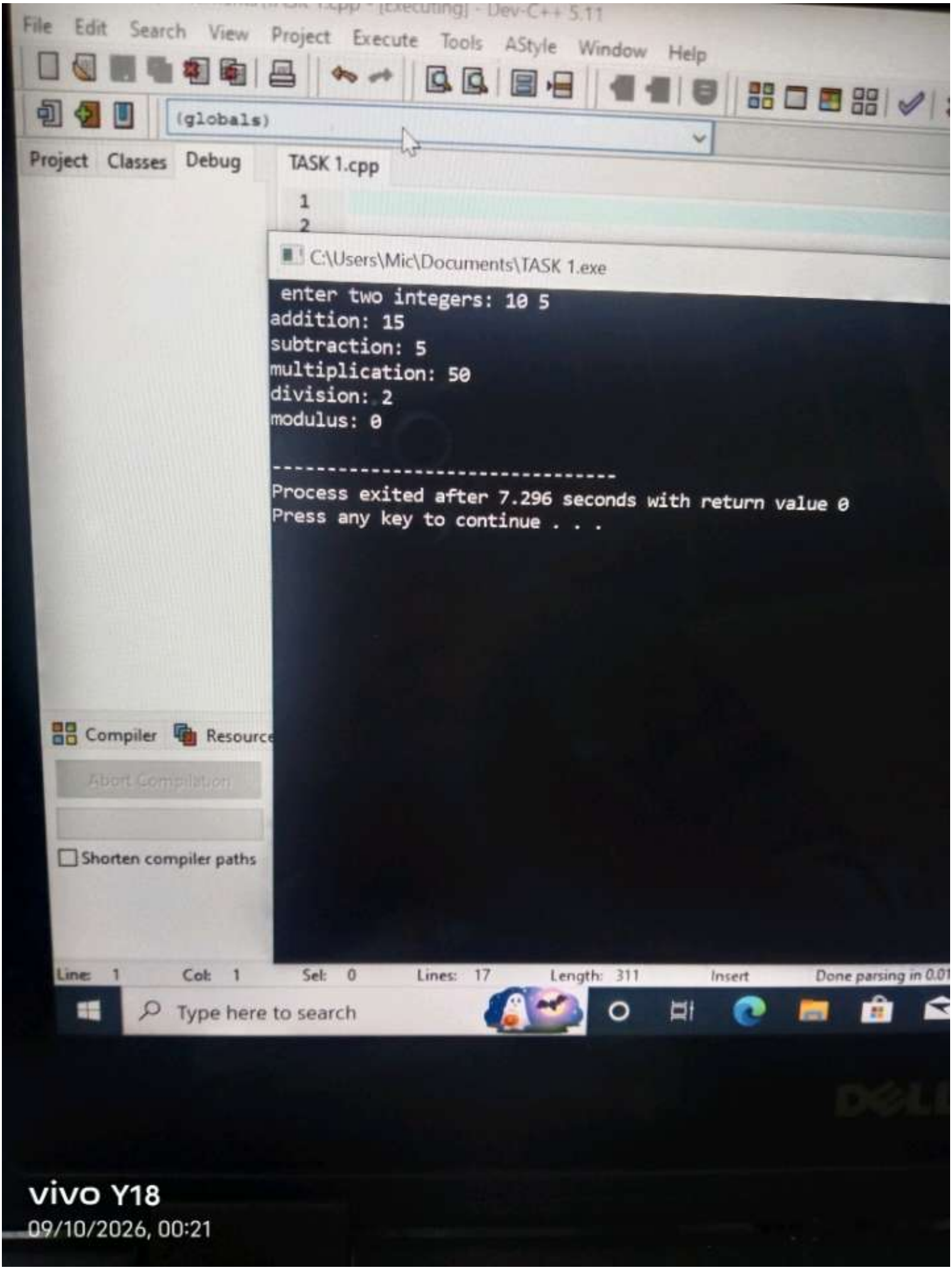
[*] TASK 1.cpp

```
1   TASK 2
2
3
4   # include <iostream>
5   using namespace std;
6   int main( ){
7
8   int a,b;
9   cout<<" enter two integers: ";
10  cin>> a>> b;
11  cout<<"addition: "<< a+b<<endl;
12  cout<<"subtraction: "<< a-b<<endl;
13  cout<<"multiplication: "<< a*b<<endl;
14  cout<<"division: "<< a/b<<endl;
15  cout<<"modulus: "<< a%b<<endl;
16  return 0;
17  }
```

Resources | Compile Log | Debug | Find Results | Close

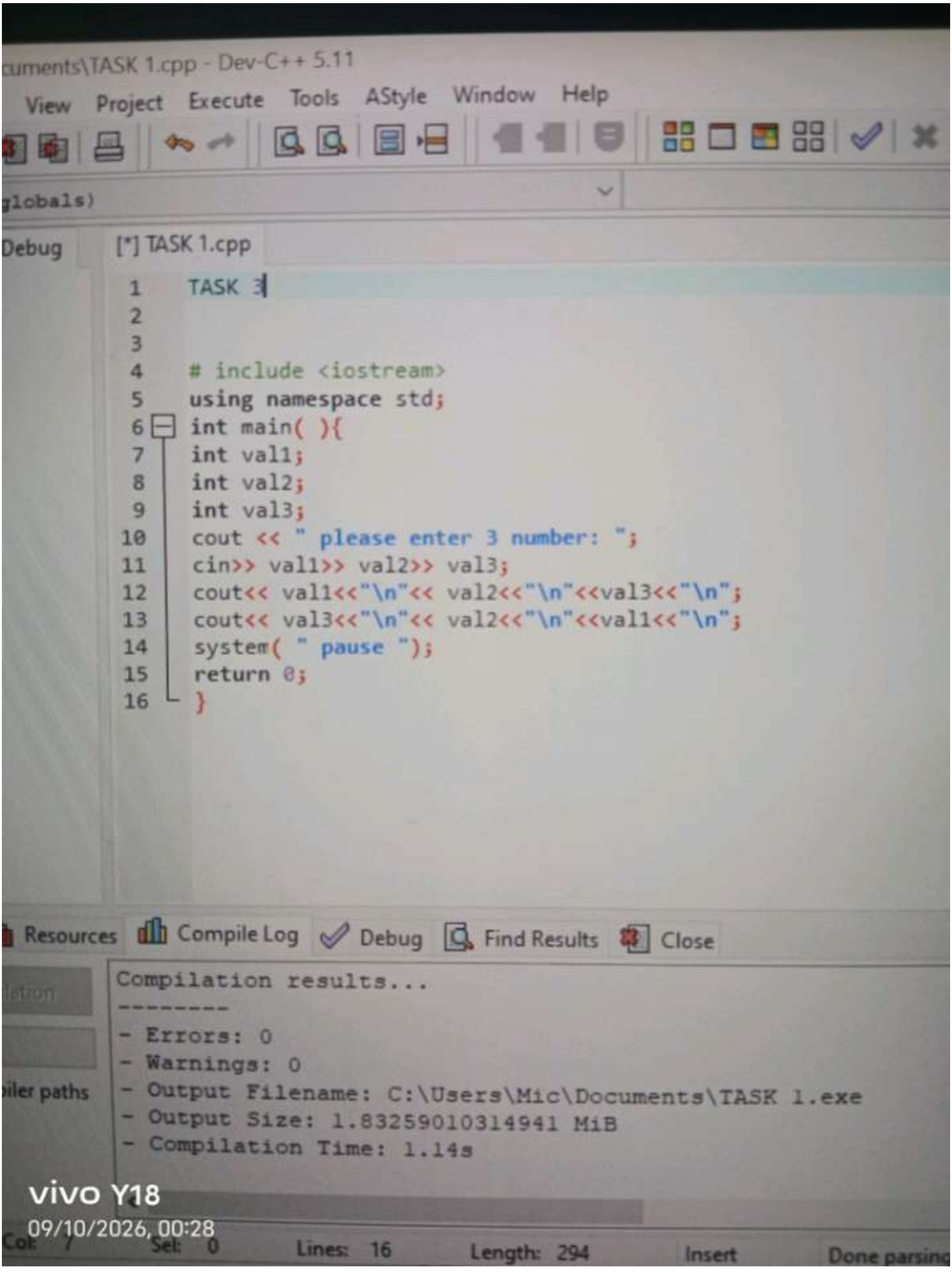
```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Mic\Documents\TASK
- Output Size: 1.833097457885 74 MiB
- Compilation Time: 1.00s
```

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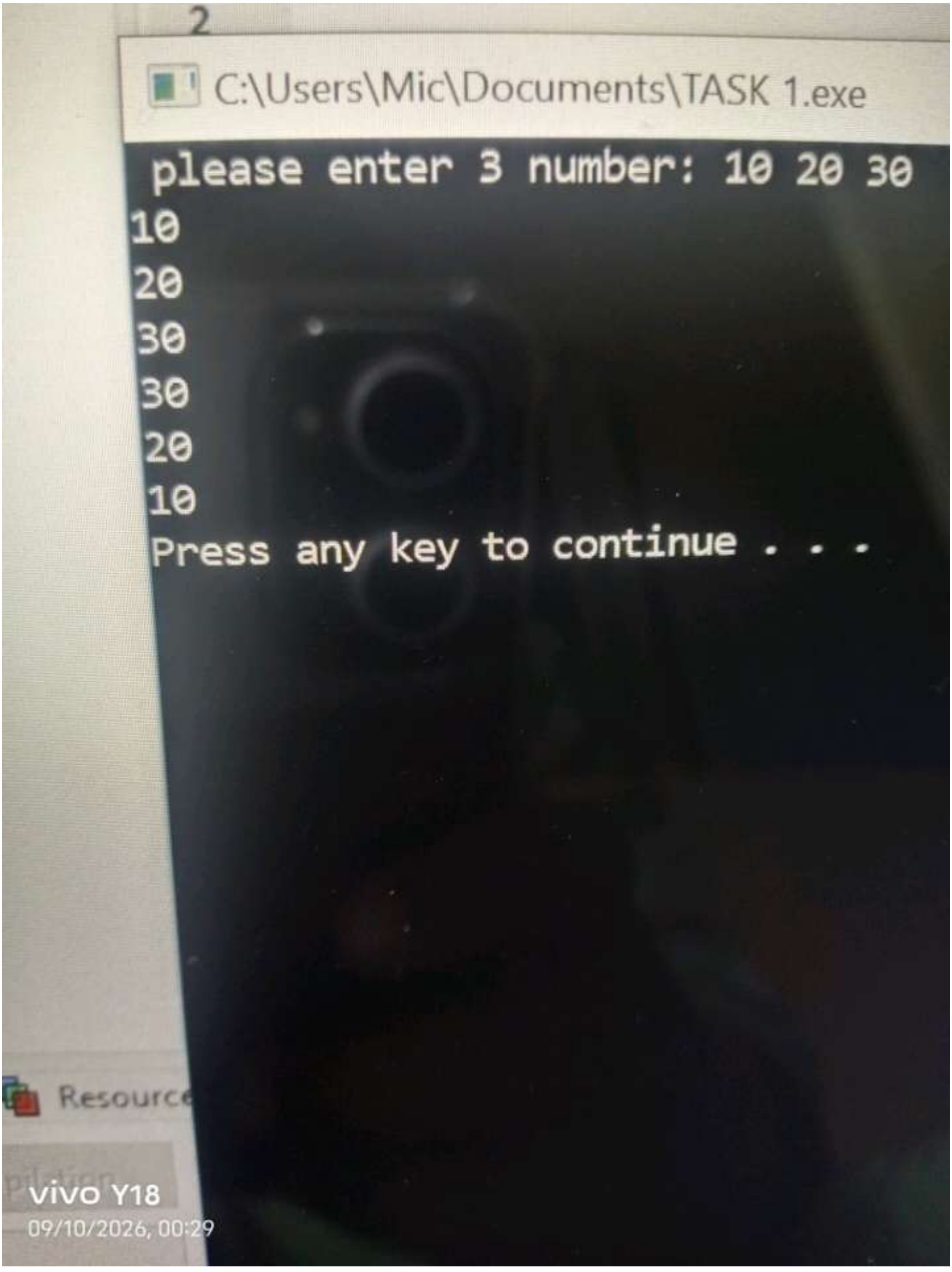



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2
C:\Users\Mic\Documents\TASK 1.exe
please enter 3 number: 10 20 30
10
20
30
30
20
10
Press any key to continue . . .
Resource
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TASK 1.cpp

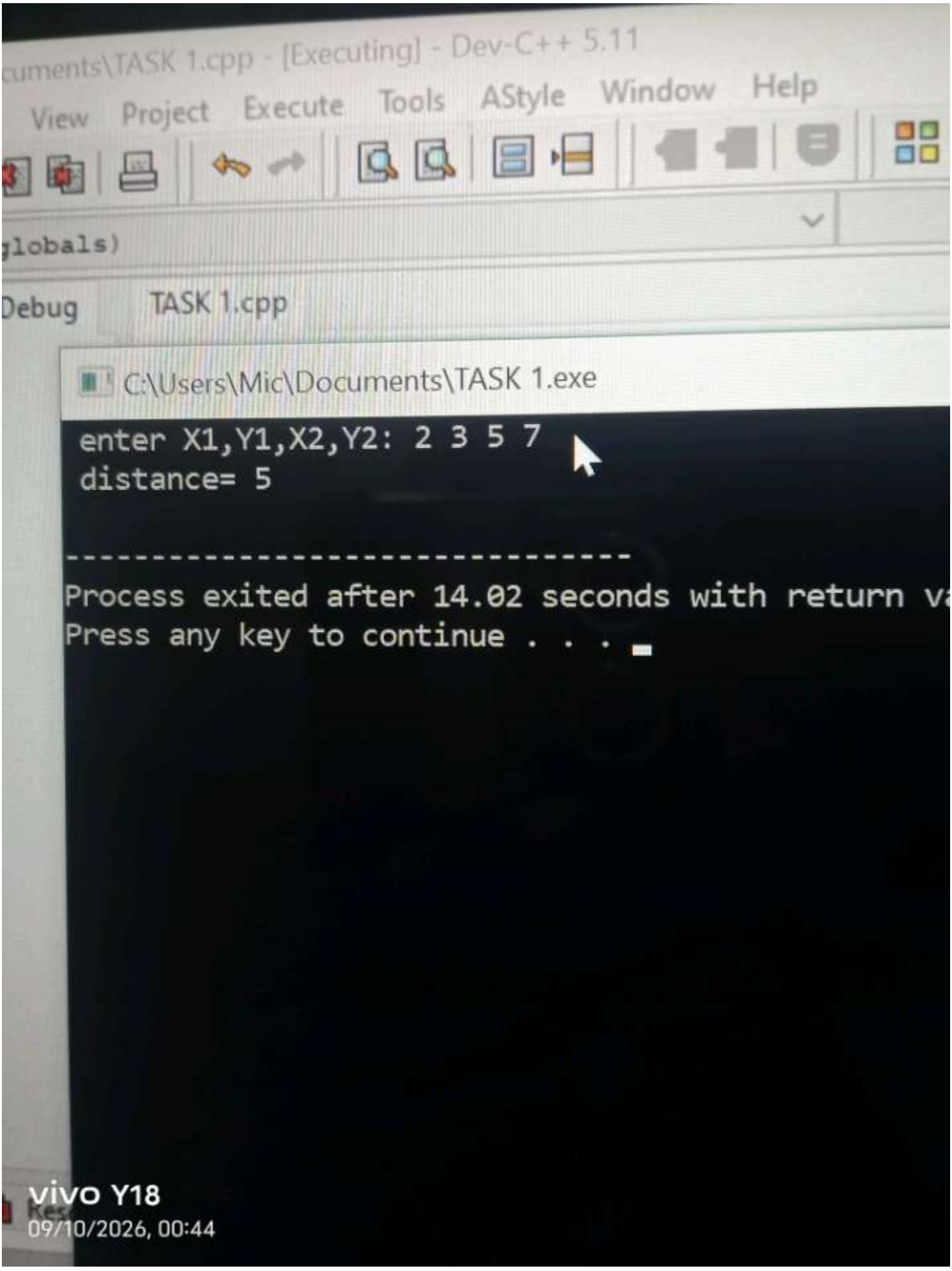
TASK 4

```cpp
# include <iostream>
# include <cmath>
using namespace std;
int main( ){
double X1,Y1,X2,Y2,d;
cout<< " enter X1,Y1,X2,Y2: ";
cin>>X1>>Y1>>X2>>Y2;
d=sqrt(pow( X2-X1,2 )+pow( Y2-Y1,2));
cout<<" distance= "<< d<< endl;
return 0;
}
```

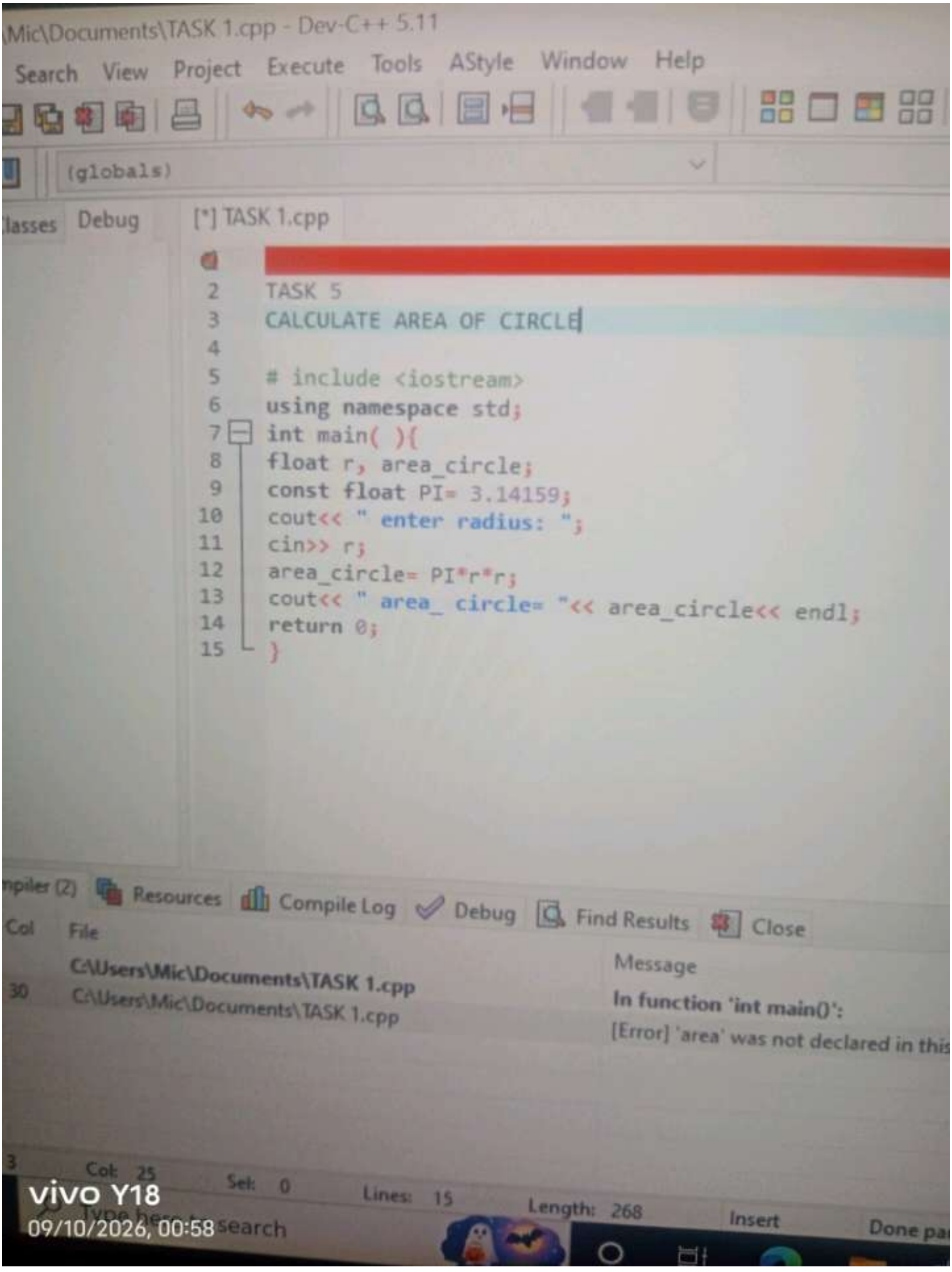
Compile Log | Debug | Find Results | Close

Compilation results...

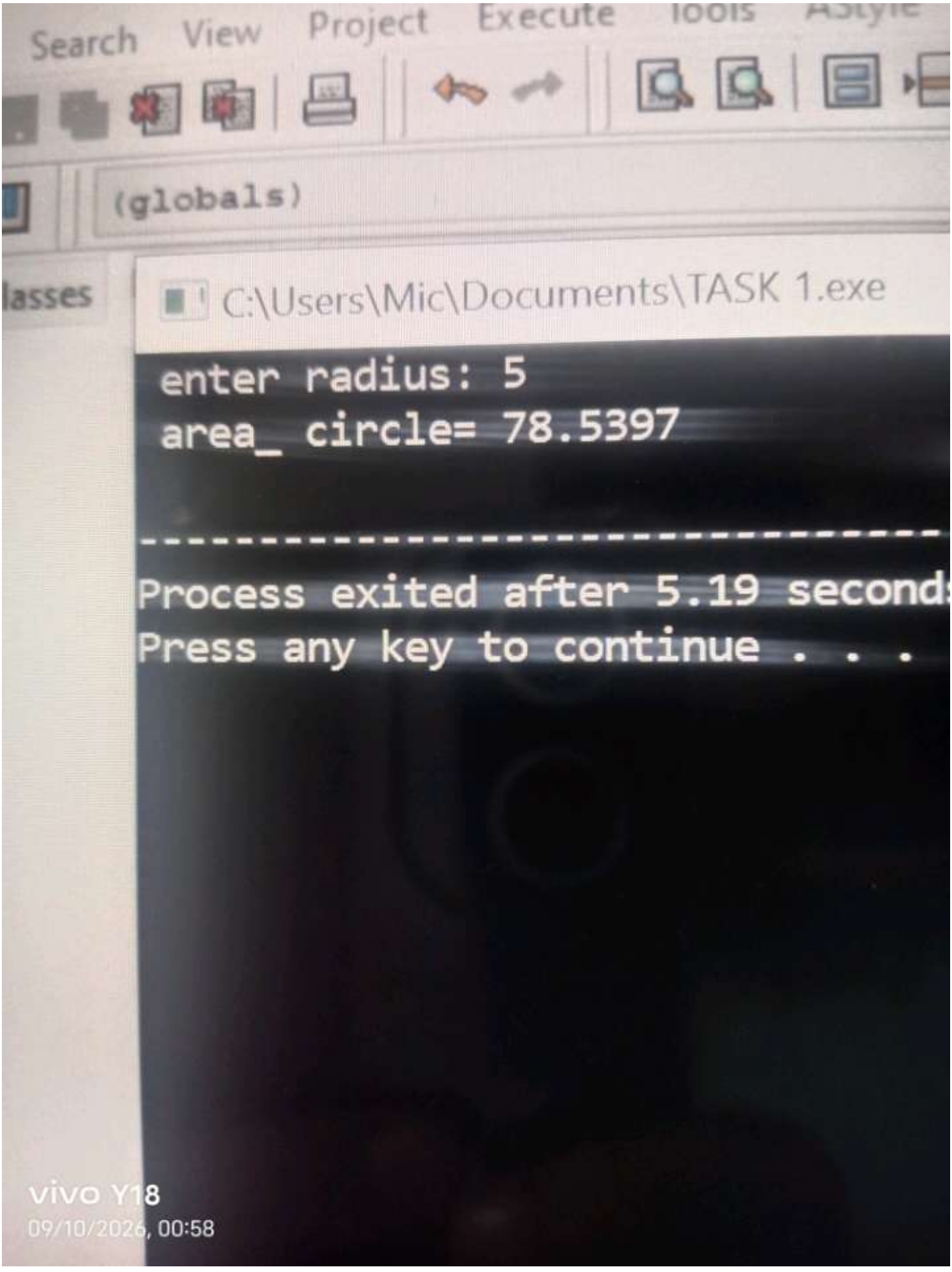
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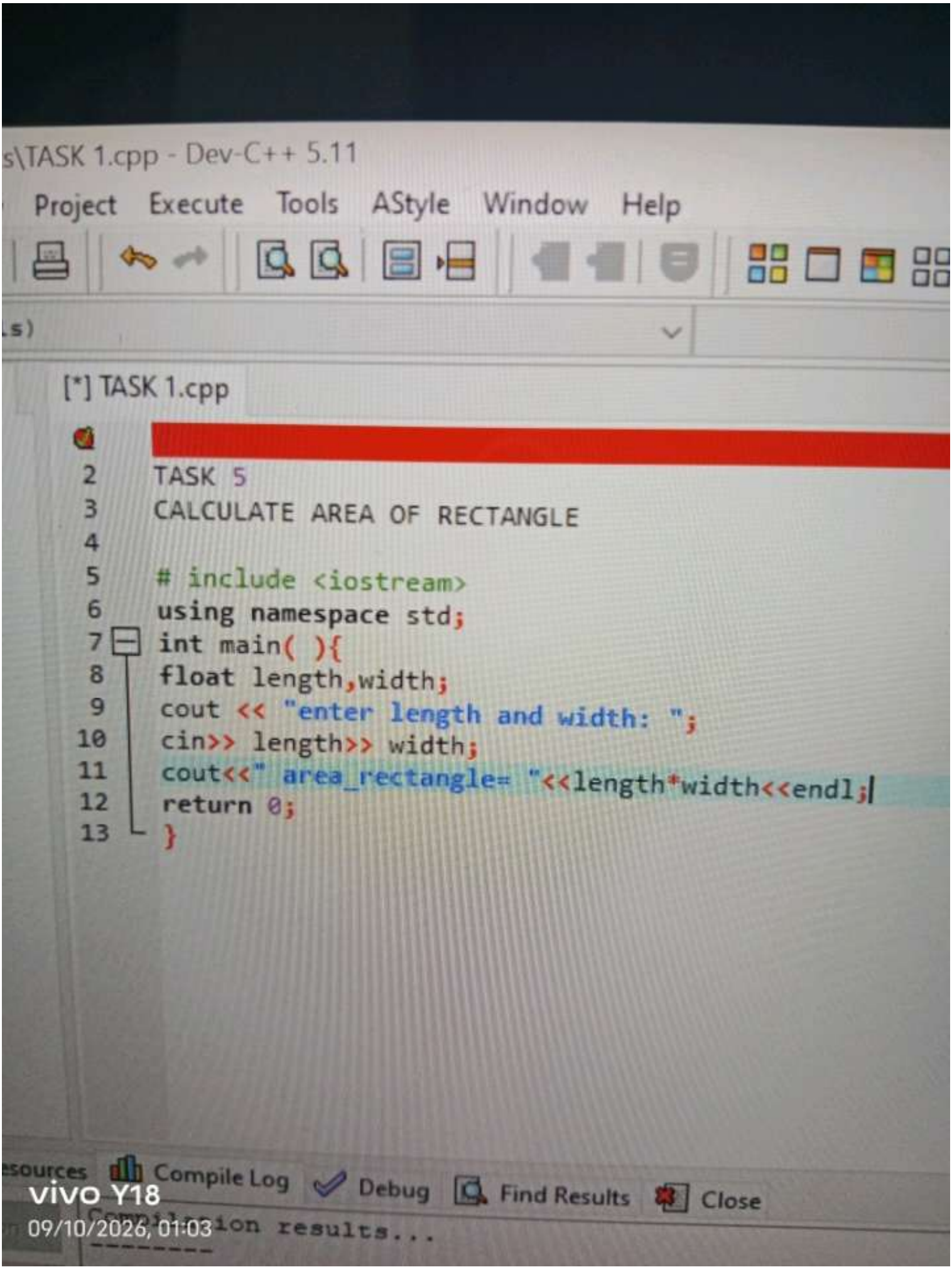


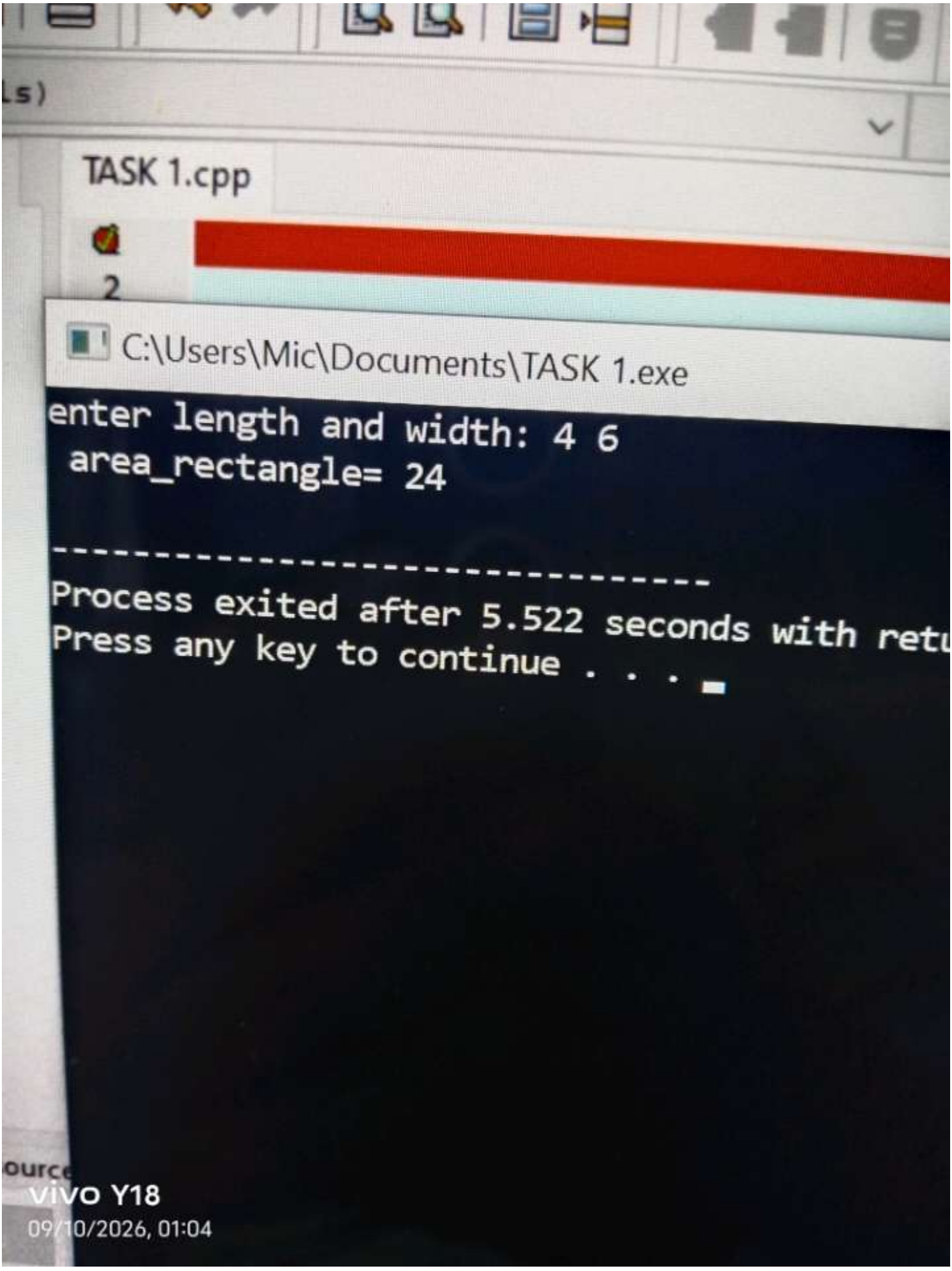

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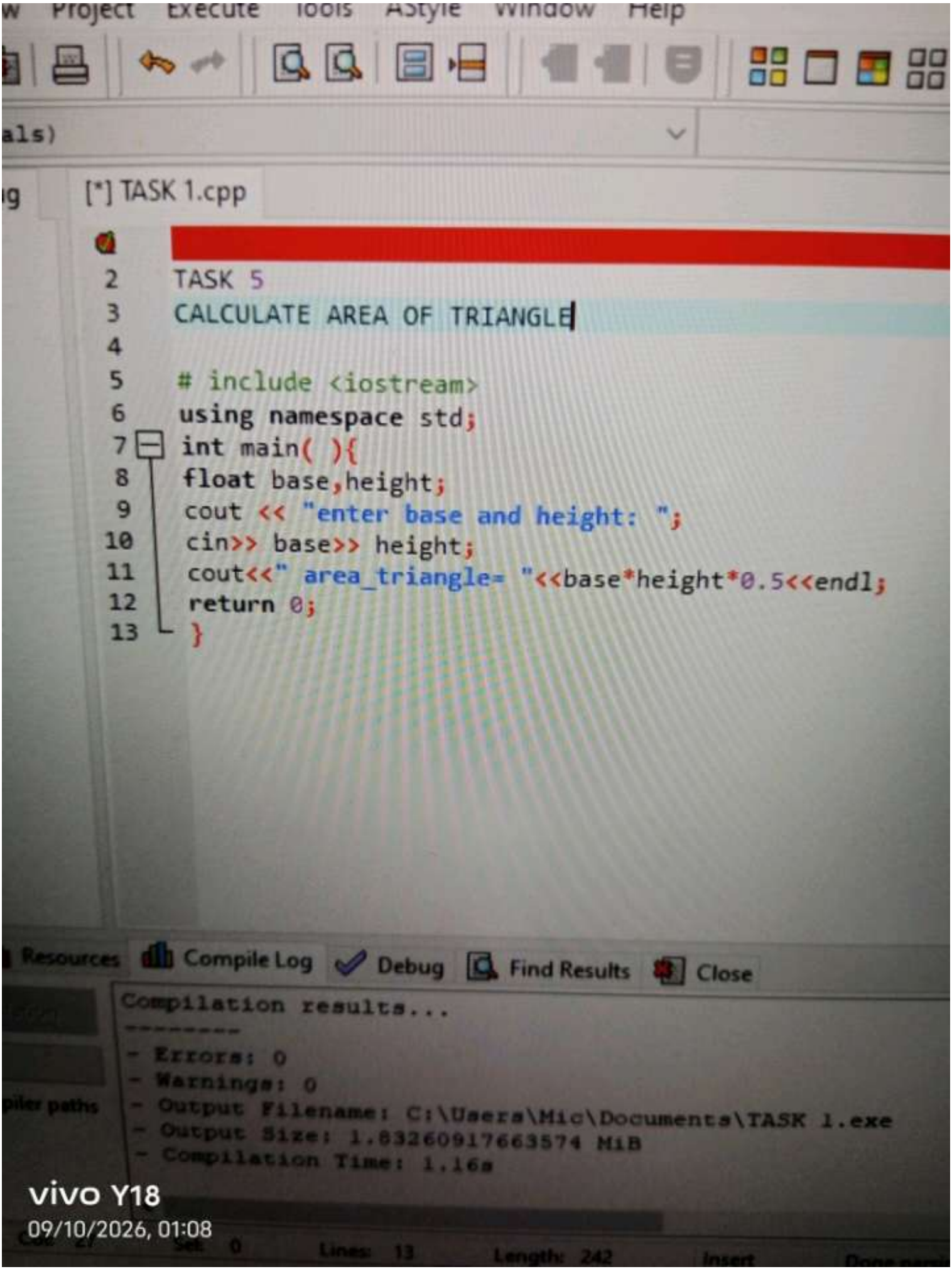
vivo Y18
09/10/2026, 00:58

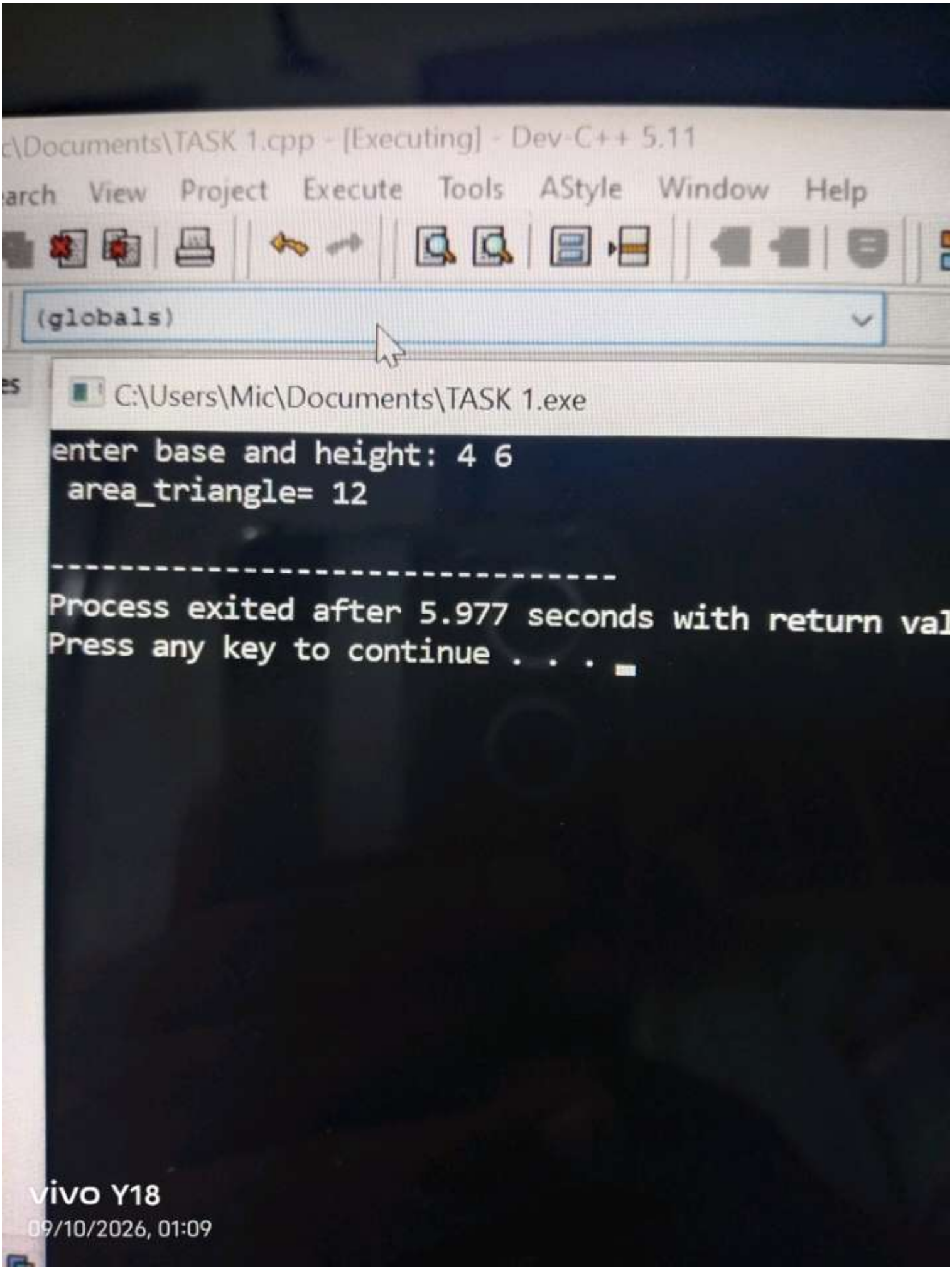

Search View Project Execute Tools AStyle
(globals)
lasses
C:\Users\Mic\Documents\TASK 1.exe
enter radius: 5
area_ circle= 78.5397
Process exited after 5.19 seconds
Press any key to continue . . .
vivo Y18
09/10/2026, 00:58


s\TASK 1.cpp - Dev-C++ 5.11
Project Execute Tools AStyle Window Help
[*] TASK 1.cpp
TASK 5
CALCULATE AREA OF RECTANGLE
# include <iostream>
using namespace std;
int main( ){
float length,width;
cout << "enter length and width: ";
cin>> length>> width;
cout<<" area_rectangle= "<<length*width<<endl;
return 0;
}
Compile Log
Debug
Find Results
Close
vivo Y18
09/10/2026, 01:03

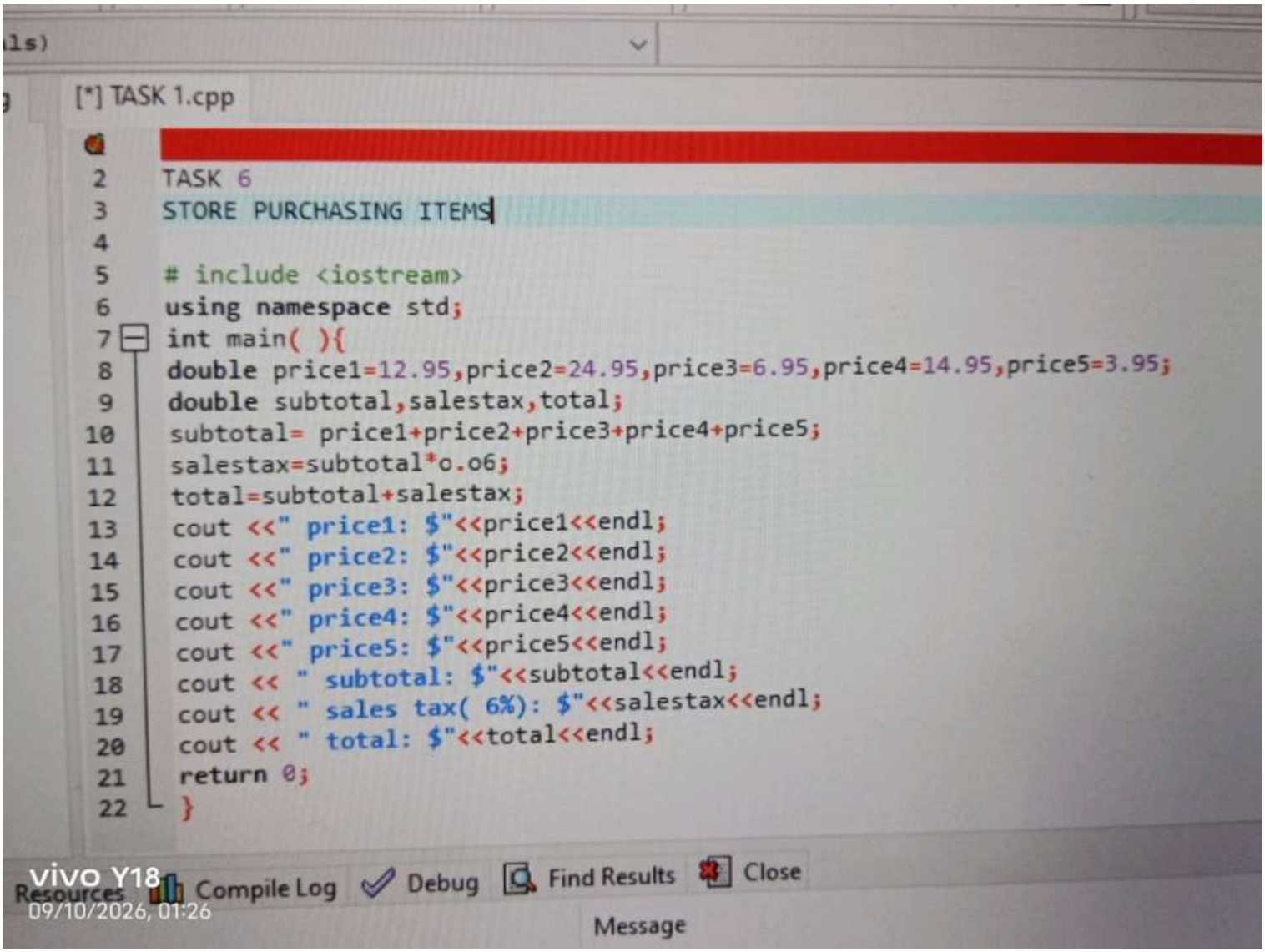



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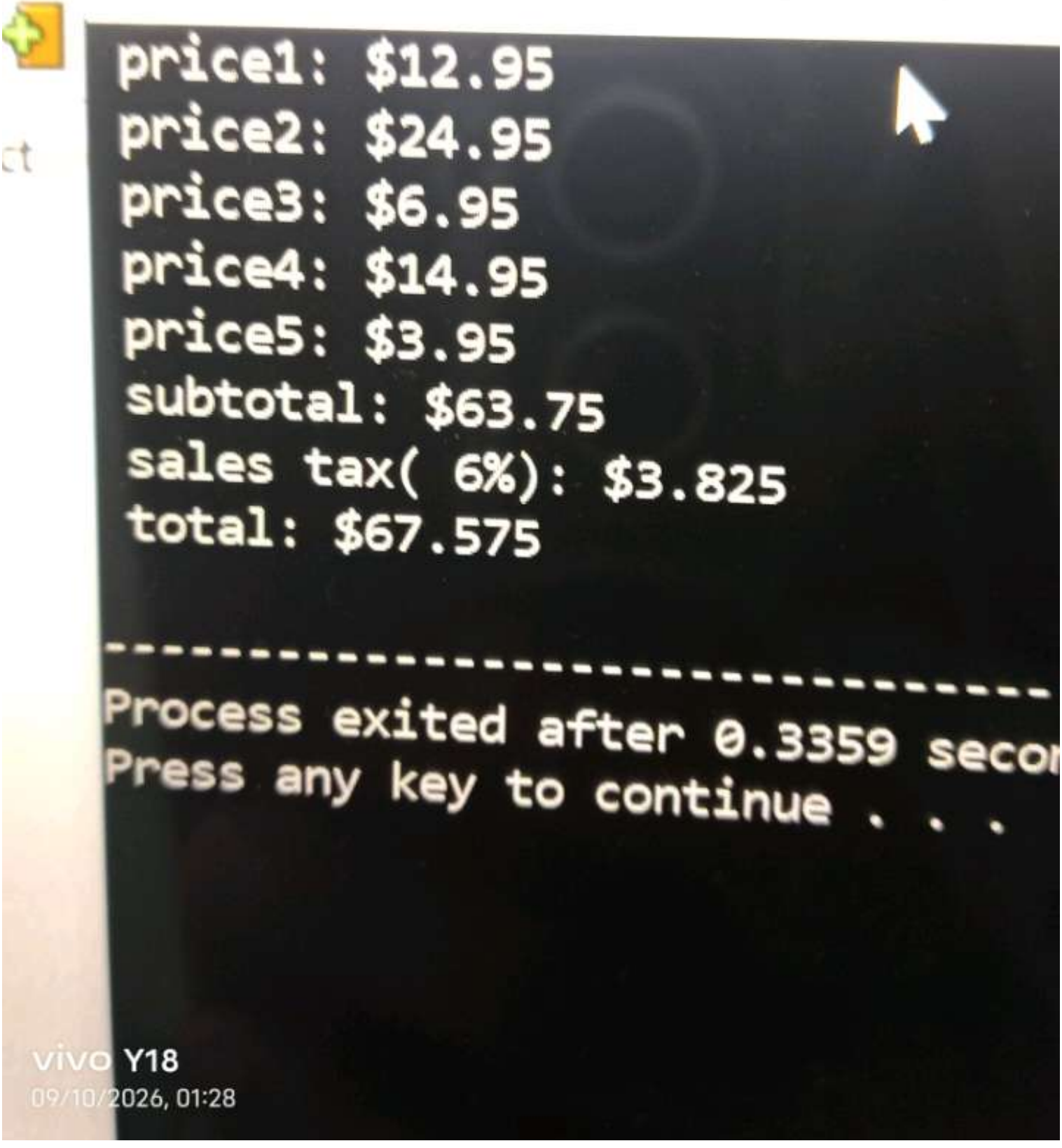

Project Execute Tools AStyle Window Help
[*] TASK 1.cpp
TASK 5
CALCULATE AREA OF TRIANGLE
# include <iostream>
using namespace std;
int main( ){
float base,height;
cout << "enter base and height: ";
cin>> base>> height;
cout<<" area_triangle= "<<base*height*0.5<<endl;
return 0;
}
Resources
Compile Log
Debug
Find Results
Close
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Mic\Documents\TASK 1.exe
- Output Size: 1.832609176635 74 MiB
- Compilation Time: 1.16s
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vivo Y18
09/10/2026, 01:09



vivo Y18
09/10/2026, 01:26




vivo Y18
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Project Execute Tools AStyle Window Help

[*] Untitled1.cpp

```
1   BONUS TASK: BREAK NUMBERS
2
3   # include <iostream>
4   using namespace std;
5
6   int main (){
7   int num;
8   cout<<" enter fivr digit integer: ";
9   cin>> num
10  int d1= num/10000;
11  int d2=( num/1000)%10;
12  int d3=( num/100)%10;
13  int d4=( num/10)%10;
14  int d5= num%10;
15  cout<<d1<<"  "<<d2<<"  "<<d3<<"  "d4<<"  "<<d5<<"  "<<endl;
16  return 0;
17  }
18
19
```

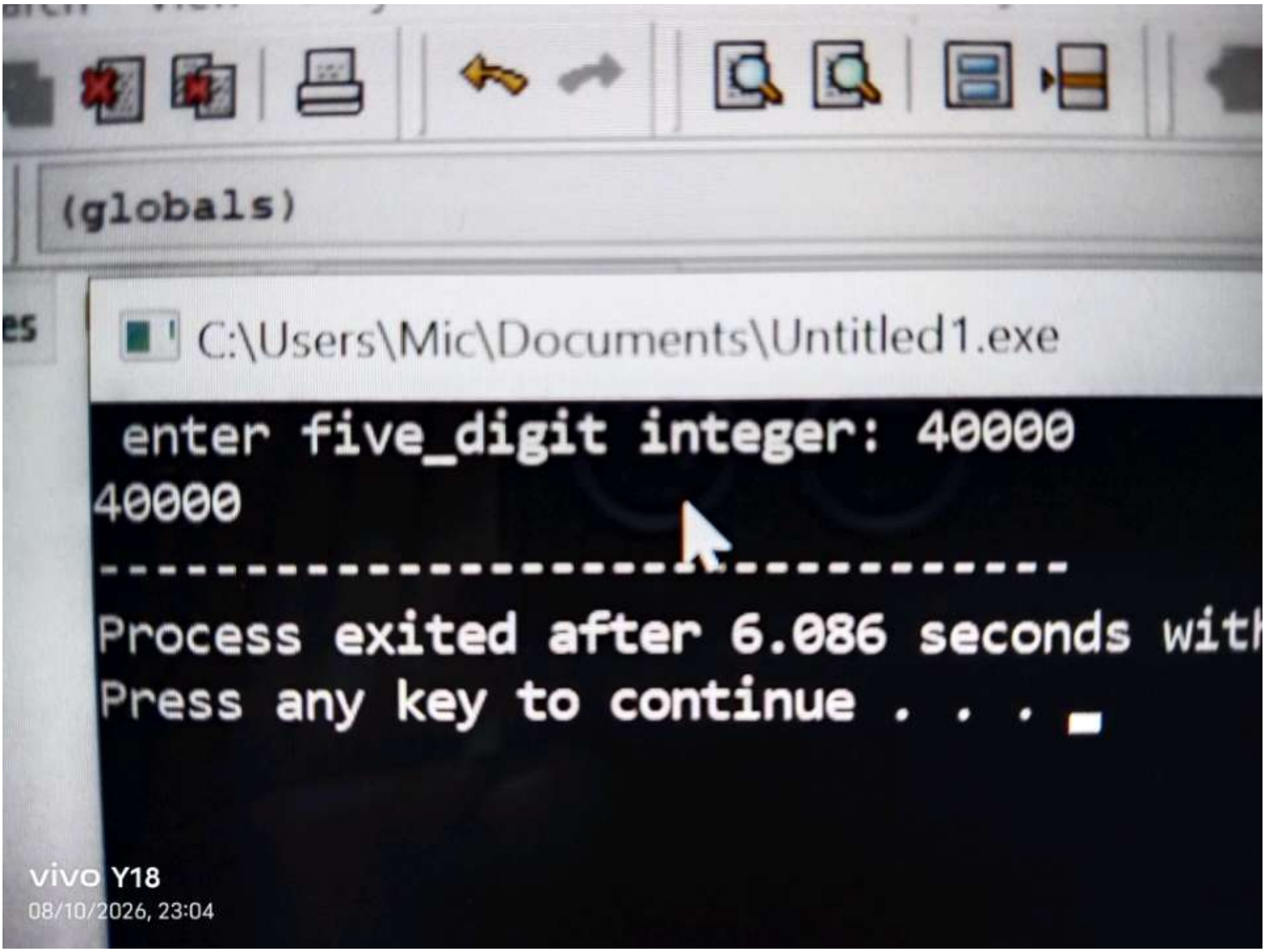
Resources Compile Log Debug Find Results

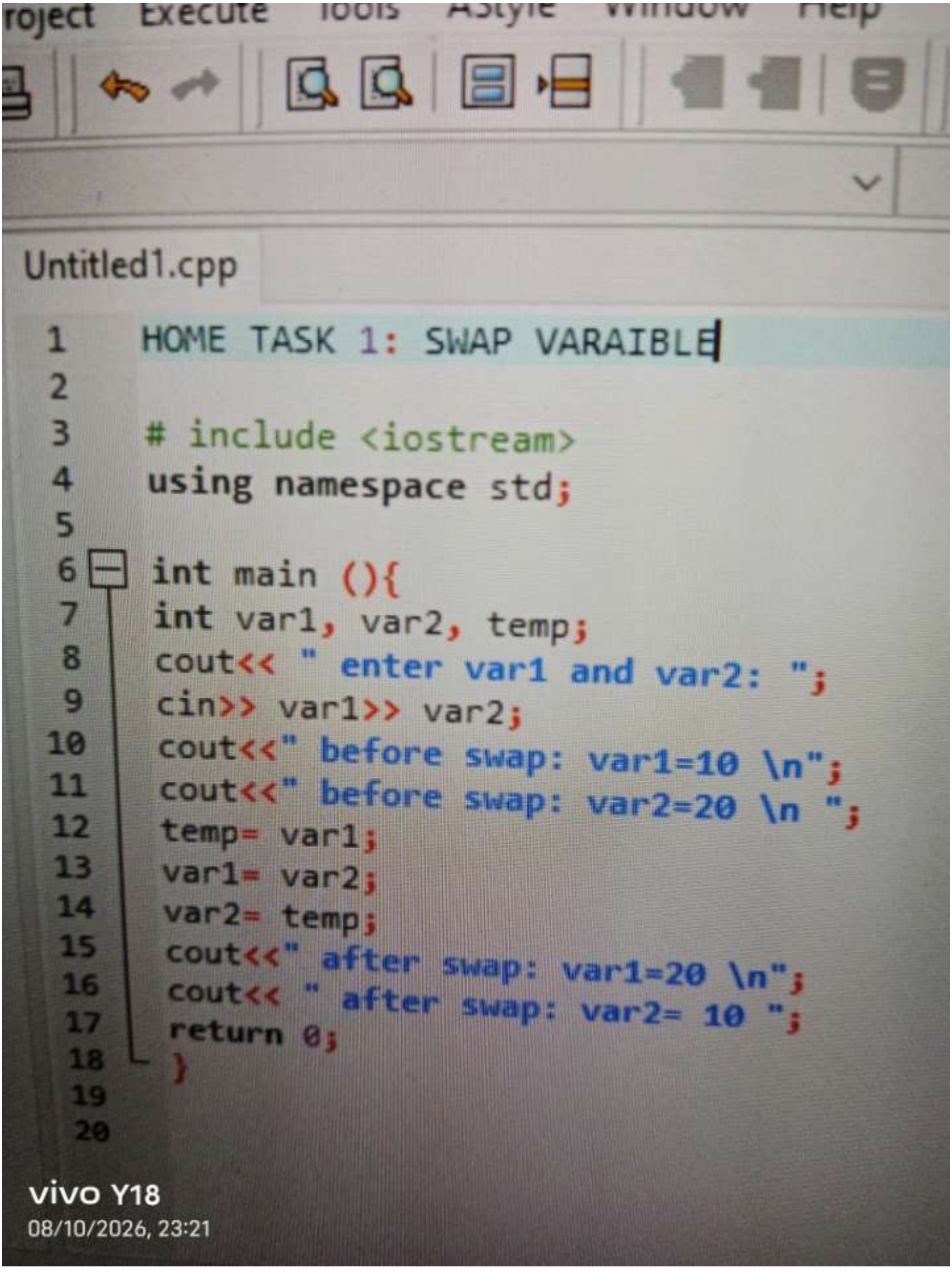
26 Sel: 0 Lines: 19 Length: 333 Insert Done parsing in 0.406

pe here to search

vivo Y18

08/10/2026, 22:54


(globals)
C:\Users\Mic\Documents\Untitled1.exe
enter five_digit integer: 40000
40000
Process exited after 6.086 seconds with
Press any key to continue . . .
vivo Y18
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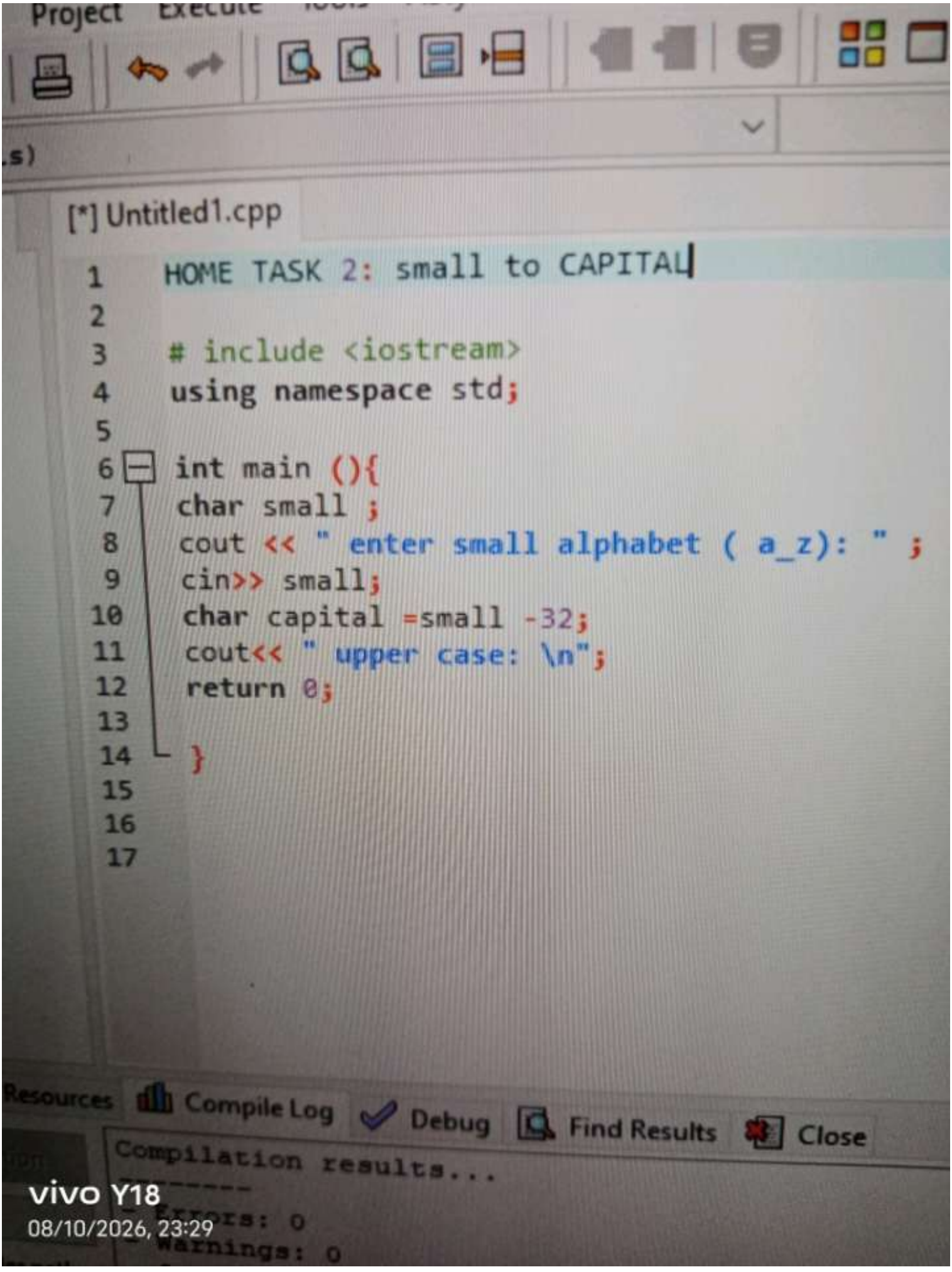

roject Execute Tools AStyle Window Help
Untitled1.cpp
1 HOME TASK 1: SWAP VARAIBLE
2
3 # include <iostream>
4 using namespace std;
5
6 int main (){
7 int var1, var2, temp;
8 cout<< " enter var1 and var2: ";
9 cin>> var1>> var2;
10 cout<<" before swap: var1=10 \n";
11 cout<<" before swap: var2=20 \n ";
12 temp= var1;
13 var1= var2;
14 var2= temp;
15 cout<<" after swap: var1=20 \n";
16 cout<< " after swap: var2= 10 ";
17 return 0;
18 }
19
20
vivo Y18
08/10/2026, 23:21

ug Untitled1.cpp

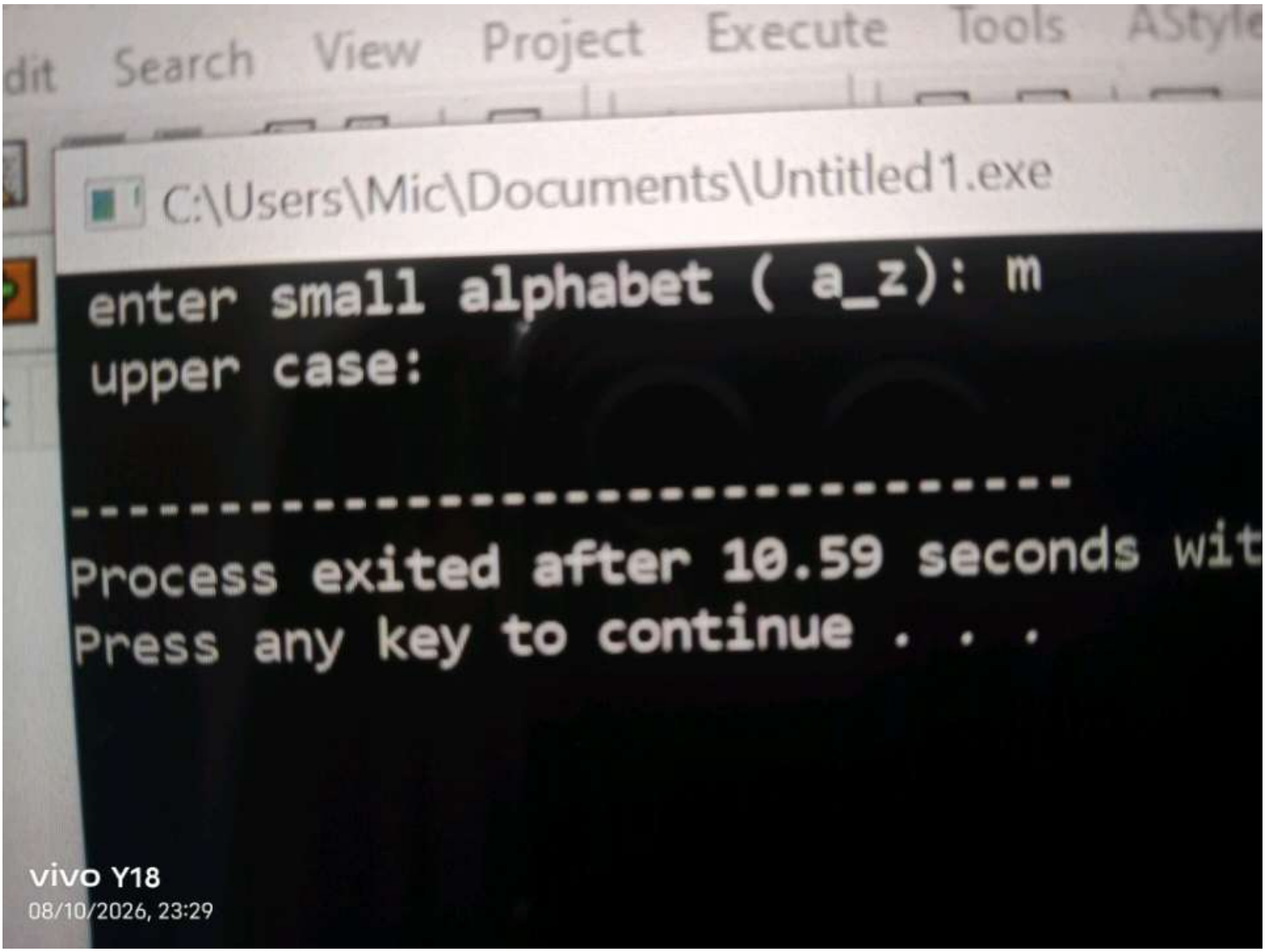
C:\Users\Mic\Documents\Untitled1.exe

```
enter var1 and var2: 10 20
before swap: var1=10
before swap: var2=20
 after swap: var1=20
after swap: var2= 10
--------------------------------
Process exited after 9.351 seconds wi
Press any key to continue . . .
```

vivo Y18
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vivo Y18
08/10/2026, 23:29


dit Search View Project Execute Tools AStyle
C:\Users\Mic\Documents\Untitled1.exe
enter small alphabet ( a_z): m
upper case:
--------------------------------
Process exited after 10.59 seconds wit
Press any key to continue . . .
vivo Y18
08/10/2026, 23:29